The Islamia University of Bahawalpur

Department of Computer Science

SOFTWARE REQUIREMENTS SPECIFICATION

(SRS DOCUMENT)

for

Thesis Vault

Version 1.0

By

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Revision History

Name	Date	Reason for changes	Version

Application Evaluation History

Comments (by committee) *include the ones given at scope time both in doc and presentation	Action Taken

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Signature _______________

Introduction

The introduction provides an overview of the ThesisVault platform and explains how the Software Requirements Specification (SRS) is structured. The document defines all functional and non-functional requirements needed to design, develop, test, and evaluate the system.

Purpose

The purpose of ThesisVault is to provide a centralized, searchable repository for final-year thesis documents. The system allows students to upload theses, supervisors to review them, and the department to publish approved theses. This SRS describes all functional and non-functional requirements of the system.

Scope

ThesisVault is a web-based platform built with Next.js and Supabase. It delivers:

1. Safe storage for thesis PDFs
2. Metadata-based search
3. Versioning support
4. Supervisor review workflow
5. Department-level publishing
   Public users can browse and download only **approved and published** theses.

Overall description

Product perspective

ThesisVault is a standalone system with no external dependencies except Supabase backend services.
It replaces manual storage methods (USB, email, Google Drive) with a single departmental repository.

Product Functions

Core functionalities:

1. Student thesis upload + metadata entry
2. Supervisor review (approve / request changes / reject)
3. Versioning (multiple uploads per thesis)
4. Publishing by admin
5. Keyword & metadata-based search
6. Secure download links for approved theses

User Classes and Characteristics

Student: Upload documents, edit metadata, resubmit revisions.
Supervisor: Review assigned theses, comment, approve/reject.
Admin: Manage users, oversee submissions, publish works.
Public/Visitor: Can view/search only published theses.

Operating environment

Web browser (Chrome, Firefox, Safari)

Hosted on Vercel (frontend)

Backend on Supabase (database + storage)

.

Design and implementation constraints

1.Must use Supabase Auth for login

2.Must use Supabase Storage for PDF files

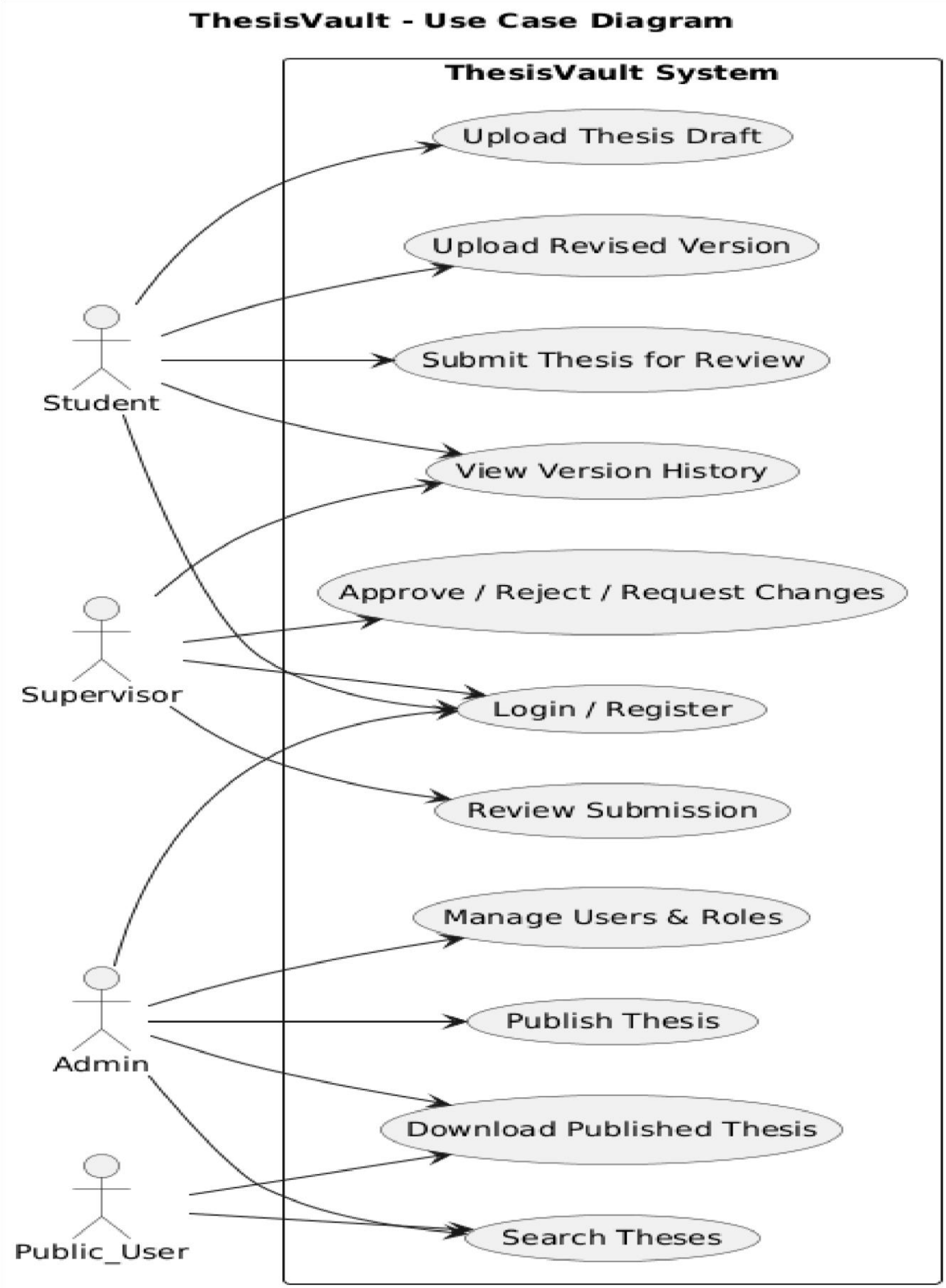
3.All metadata stored in PostgreSQL

4.Role-based access enforced via Supabase RLS policies

Requirement identifying technique

Requirement identifying techniques were applied to capture the functional and non-functional needs of ThesisVault, considering its academic nature and the workflow of thesis uploading, reviewing, and publishing. Since ThesisVault is a multi-actor system involving Students, Supervisors, and Admins, techniques focusing on user interaction and event flow were prioritized. The following techniques were used to identify system requirements:

Use case diagram


ThesisVault - Use Case Diagram
ThesisVault System
Upload Thesis Draft
Upload Revised Version
Submit Thesis for Review
View Version History
Approve / Reject / Request Changes
Login / Register
Review Submission
Manage Users & Roles
Publish Thesis
Download Published Thesis
Search Theses
Student
Supervisor
Admin
Public_User

The Use Case Diagram illustrates the functional scope of the ThesisVault system by identifying the system actors and the interactions they have with the system. It represents the major functionalities provided by the system from the user's perspective and defines the boundaries of the system.

The diagram includes three primary actors: **Student**, **Supervisor**, and **Administrator**. Each actor interacts with the system based on their assigned role and responsibilities. The use cases show how thesis documents are uploaded, reviewed, approved, and published within the system.

This diagram helps in understanding system requirements, identifying functional boundaries, and ensuring that all user roles are supported by appropriate system functionalities.

Use case description

These table below indicate a comprehensive use case filled in with the Thesis Management System.

Use Case ID:	UC-1
Use Case Name:	User Login & Authentication
Actors:	Student, Supervisor, Admin

Preconditions:	User has a valid account.
Postconditions:	User is logged into the system.
Normal Flow:	1. User enters email and password. 2. System verifies credentials through Supabase Auth. 3. System redirects to respective dashboard based on role.

Use Case ID:	UC-2
Use Case Name:	Upload Thesis Document
Actors:	Student
Preconditions:	User is authenticated.
Postconditions:	Thesis is saved as a draft.

Normal Flow:	1. Student opens “Upload Thesis” form. 2. Student enters metadata (title, abstract, keywords, year, program, supervisor). 3. Student uploads PDF file. 4. System validates file and stores it in Supabase Storage. 5. System creates thesis draft record.

Use Case ID:	UC-3
Use Case Name:	Submit Thesis for Review
Actors:	Student
Preconditions:	Thesis is in draft state.
Postconditions:	Supervisor sees the thesis under pending reviews.

Normal Flow:	1. Student selects a draft thesis. 2. Student clicks “Submit for Review.” 3. System updates status to “Submitted.” 4. System notifies assigned supervisor.

Use Case ID:	UC-4
Use Case Name:	Review Thesis Submission
Actors:	Supervisor
Preconditions:	A thesis is submitted for review.
Postconditions:	Thesis status updated accordingly.

Normal Flow:	1. Supervisor opens the submission. 2. Supervisor reads metadata and PDF. 3. Supervisor selects “Approve,” “Reject,” or “Request Changes.” 4. System records decision and comment. 5. System notifies the student.

Functional Requirements

FR ID:	FR-1
FR Name:	User Registration & Login
Requirements:	The system shall allow users (Student, Supervisor, and Administrator) to register using email and password and log in securely.
Source:	Stakeholder meetings and project scope definition.
Rationale:	User authentication is essential to provide role-based access and protect thesis data.

FR ID:	FR-2
FR Name:	Role-Based Access Control

Requirements:	The system shall assign a role (Student, Supervisor, or Administrator) to each registered user and restrict system functionality based on the assigned role.
Source:	Stakeholder meetings and project scope definition.
Rationale:	Role-based access ensures data confidentiality and enforces proper authorization.

FR ID:	FR-3
FR Name:	Thesis Draft Creation
Requirements:	The system shall allow students to create a thesis draft before submitting it for review.
Source:	Academic workflow requirements.
Rationale:	Draft creation allows students to prepare and manage thesis content incrementally.

FR ID:	FR-4
FR Name:	Thesis Upload and Metadata Entry
Requirements:	The system shall allow students to upload thesis documents in PDF format and enter mandatory metadata including title, abstract, year, program, keywords, and supervisor.
Source:	Department thesis submission guidelines.

Rationale:	Metadata improves organization, search ability, and academic record keeping.

FR ID:	FR-5
FR Name:	Thesis Submission for Review
Requirements:	The system shall allow students to submit a thesis draft for supervisor review and update its status to “Submitted”.
Source:	Thesis review workflow requirements.
Rationale:	Formal submission initiates the review and approval process.

FR ID:	FR-6
FR Name:	Supervisor Review and Feedback
Requirements:	The system shall allow supervisors to review submitted theses and approve, reject, or request changes with comments.
Source:	Supervisor feedback process.
Rationale:	Supervisor evaluation is essential for academic quality assurance.

Non-Functional Requirements

NFR-1 Performance

1. System should return search results within 1 second for typical data load.
2. File upload shall complete within reasonable time based on file size and connection.

NFR-2 Security

1. PDF files must be stored in protected storage buckets.
2. RLS (Row Level Security) must enforce access rules per role. Only published theses may be publicly visible.
3. Passwords secured through Supabase Auth (hashed & salted).

NFR-3 Reliability

1. System shall keep all previous versions without overwriting.
2. System shall recover from failed uploads with appropriate user messages. System shall be available 24/7 (excluding maintenance).

NFR-4 Usability

1. UI must be clean, simple, and responsive.
2. Upload flow should require minimal steps.
3. Dashboard should clearly display status indicators (Draft, Submitted, Review, Approved, Published).

NFR-5 Maintainability

1. Code structure shall follow modular Next.js page and API routing.
2. Database schema shall adhere to 3rd Normal Form.
3. Business logic must be clearly separated from presentation.

NFR-6 Scalability

1. System must support growth of thesis records without redesign.
2. Search must remain fast as dataset grows.

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